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Instruction Set and Programming with 8085

GATE ECE Microprocessors · Previous Year Questions with Answer Key

36	29	1988 – 2021	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	16 questions
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Answer key	at the end

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Marks 1 16 questions

Q1. GATE ECE 2020

In an 8085 microprocessor, the number of address lines required to access a 16 K bytes memory bank is

____ .

Open solution

Q2. GATE ECE 2017 Set 1

The clock frequency of an 8085 microprocessor is 5 MHz. If the time required to execute an instruction is 1.4 μs, then the number of T-states needed for executing the instruction is

(A) 1 (B) 6 (C) 7 (D) 8

Open solution

Q3. GATE ECE 2016 Set 3

In an 8085 microprocessor, the contents of the accumulator and the carry flag are A7 (in hex) and 0, respectively. If the instruction RLC is executed, then the contents of the accumulator (in hex) and the carry flag, respectively, will be

(A) 4E and 0 (B) 4E and 1 (C) 4F and 0 (D) 4F and 1

Open solution

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Q4. GATE ECE 2015 Set 3

Which one of the following 8085 microprocessor programs correctly calculates the product of two 8-bit numbers stored in registers B and C?

Open solution

(A)

```
MVI A, 00H
JNZ LOOP
CMP C
LOOP DCR B
HLT
```

(B)

```
MVI A, 00H
CMP C
LOOP DCR B
JNZ LOOP
HLT
```

(C)

```
MVI A, 00H
LOOP ADD C
DCR B
JNZ LOOP
HLT
```

(D)

```
MVI A, 00H
ADD C
JNZ LOOP
LOOP DCR B
HLT
```

Q5. GATE ECE 2015 Set 2

In an 8085 microprocessor, which one of the following instructions changes the content of the accumulator?

(A) MOV B, M (B) PCHL (C) RNZ (D) SBI BEH

Open solution

Q6. GATE ECE 2015 Set 1

In an 8085 microprocessor, the shift registers which store the result of an addition and the overflew bit are, respectively.

(A) B & F (B) A & F (C) H & F (D) A & C

Open solution

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Q7. GATE ECE 2013

For 8085 microprocessor, the following program is executed.

```
      MVI A, 05H;
      MVI B, 05H;
PTR:  ADD B;
      DCR B;
      JNZ PTR;
      ADI 03H;
      HLT;
```

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At the end of program, accumulator contains

(A) 17 H (B) 20 H (C) 23 H (D) 05 H

Open solution

Q8. GATE ECE 1998

An instruction used to set the carry flag in a computer can be classified as

(A) data transfer (B) arithmetic (C) logical (D) program control

Open solution

Q9. GATE ECE 1997

The following instructions have been executed by an 8085 μP

ADDRESS (HEX)	INSTRUCTION
6010	LXI H, 8A, 79 H
6013	MOV A, L
6015	ADDH
6016	DAA
6017	MOV H, A
6018	PCHL

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From which address will be next instruction be fetched?

(A) 6019 (B) 0379

(C) 6979 (D) None of the above

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Q10. GATE ECE 1996

The following sequence of instructions are executed by an 8085 microprocessor:
1000: LXI SP< 27FF
1003: CALL 1006
1006: POP H
The contents of the stack pointer (SP) and the HL register pair on completion or execution of these instructions are.

(A) SP= 27FF, HL = 1003
(B) SP= 27FD, HL = 1003
(C) SP= 27FF, HL = 1006
(D) SP= 27FD, HL = 1006

Open solution

Q11. GATE ECE 1996

The total number of memory accesses involved (inclusive of the op-code fetch) when an 8085 processor executes, the instruction LDA 2003 is

(A) 1
(B) 2
(C) 3
(D) 4

Open solution

Q12. GATE ECE 1995

An 'Assembler' for a microprocessor a used for

(A) assembly of processors in a production lines.
(B) creation of new programs using different modules.
(C) translation of a program from assembly language to machine language.
(D) translation of a higher level language into English text.

Open solution

Q13. GATE ECE 1993

In a microprocessor, the register which holds the address of the next instruction to be fetched is

(A) Accumulator
(B) Program Counter
(C) Stack Pointer
(D) Instruction Register

Open solution

Q14. GATE ECE 1992

The following program is run on 8085 microprocessor:

Memory Address in hex	Instruction
2000	LXI SP, 1000
2003	PUSH H
2004	PUSH D
2005	CALL 2050
2008	POP H
2009	HLT

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At the completion of execution of the program, the program Counter of the 8085 contains _____ and the Stack Pointer contains _____.

Open solution

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Q15. GATE ECE 1989

In a microprocessor system, the stack is used for

(A) storing the program return address whenever a sub-routine jump instruction is executed.

(B) transmitting and receiving input-output data.

(C) storing all important CPU register contents whenever an interrupt is to be serviced.

(D) storing program instructions for interrupt service routines.

Open solution

Q16. GATE ECE 1988

In register index addressing mode the effective address is given by

(A) The index register Value.

(B) The sum of the index register Value and the operand.

(C) The operand.

(D) The difference of the index register Value and the operand.

Open solution

Marks 2 18 questions

Q17. GATE ECE 2021

The content of the registers are $R_1 = 25\text{H}, R_2 = 30\text{H}$ and $R_3 = 40\text{H}$. The following machine instructions are executed.

$\text{PUSH}\,\{R_1\}$
$\text{PUSH}\,\{R_2\}$
$\text{PUSH}\,\{R_3\}$
$\text{POP}\,\{R_1\}$
$\text{POP}\,\{R_2\}$
$\text{POP}\,\{R_3\}$

After execution, the content of registers R_1, R_2, R_3 are

(A) $R_1 = 40\text{H}, R_2 = 30\text{H}, R_3 = 25\text{H}$

(B) $R_1 = 40\text{H}, R_2 = 25\text{H}, R_3 = 30\text{H}$

(C) $R_1 = 25\text{H}, R_2 = 30\text{H}, R_3 = 40\text{H}$

(D) $R_1 = 30\text{H}, R_2 = 40\text{H}, R_3 = 25\text{H}$

Open solution

Q18. GATE ECE 2017 Set 1

The following five instructions were executed on an 8085 microprocessor.
MVI A, 33H
MVI B, 78H
ADD B
CMA
ANI 32H
The Accumulator value immediately after the execution of the fifth instruction is

(A) 00H (B) 10H (C) 11H (D) 32H

Open solution

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Q19. GATE ECE 2016 Set 1

In an 8085 system, a PUSH operation requires more clock cycles than a POP operation. Which one of the following options is the correct reason for this?

(A) For POP, the data transceivers remain in the same direction as for instruction fetch (memory to processor), whereas for PUSH their direction has to be reversed.

(B) Memory write operations are slower than memory read operations in an 8085 based system.

(C) The stack pointer needs to be pre-decremented before writing registers in a PUSH, whereas a POP operation uses the address already in the stack pointer.

(D) Order of registers has to be interchanged for a PUSH operation, whereas POP uses their natural order.

Open solution

Q20. GATE ECE 2014 Set 4

An 8085 microprocessor executes "STA 1234H" with starting address location 1FFEH (STA copies the contents of the Accumulator to the 16-bit address location). While the instruction is fetched and executed, the sequence of values written at the address pins A_{15} – A_8 is

(A) 1FH, 1FH, 20H, 12H

(B) 1FH, FEH, 1FH, FFH, 12H

(C) 1FH, 1FH, 12H, 12H

(D) 1FH, 1FH, 12H, 20H, 12H

Open solution

Q21. GATE ECE 2011

An 8085 assembly language program is given below. Assume that the carry flag is initially unset. The content of the accumulator after the execution of the program is

```
MVI A,07H
RLC
MOV B,A
RLC
RLC
ADD B
RRC
```

(A) 8C H

(B) 64 H

(C) 23 H

(D) 15 H

Open solution

Q22. GATE ECE 2010

For the 8085 assembly language program given below, the content of the accumulator after the executions of the program is

```
3000 MVI A, 45H
3002 MOV B, A
3003 STC
3004 CMC
3005 RAR
3006 XRA B
```

(A) OOH

(B) 45H

(C) 67H

(D) E7H

Open solution

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Q23. GATE ECE 2008

Open solution

An 8085 executes the following instructions
2710 LXI H, 30A0H
DAD H
PCHL
All addresses and constants are in Hex. Let PC be the contents of the program counter and HL be the contents of the HL register pair just after executing PCHL.
Which of the following statements is correct.

(A) PC 2715H HL 30A0H
(B) PC 30A0H HL 2715H
(C) PC 6140H HL 6140H
(D) PC 6140H HL 2715H

Q24. GATE ECE 2007

Open solution

An 8085 assembly language program is given below.
Line 1: MVI A, B5H
2: MVI B, 0EH
3: XRI 69H
4: ADD B
5: ANI 9BH
6: CPI 9FH
7: STA 3010H
8: HLT
The contents of the accumulator just after execution of the ADD instruction in line 4 will be

(A) C3H
(B) EAH
(C) DCH
(D) 69H

Q25. GATE ECE 2007

Open solution

An 8085 assembly language program is given below.
Line 1: MVI A, B5H
2: MVI B, 0EH
3: XRI 69H
4: ADD B
5: ANI 9BH
6: CPI 9FH
7: STA 3010H
8: HLT
After execution of line of the program, the status of the CY and Z flags will be

(A) CY= 0, Z=0
(B) CY= 0, Z=1
(C) CY= 1, Z=0
(D) CY= 1, Z=1

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Q26. GATE ECE 2006

Open solution

Following is the segment of a 8085 assembly Language program:
LXI SP, EFFF H
CALL 3000 H
3000H: LXI H, 3CF4H
PUSH PSW
SPHL
POP PSW
RET
On completion of RET execution, the contents of SP is

(A) 3CF0H (B) 3CF8H (C) EFFDH (D) EFFFH

Q27. GATE ECE 2005

Open solution

Consider an 8085 microprocessor system

The following program starts at locaton 0100H.
LXI SP, 00FF
LXI H, 0107
MVI A, 20H
SUB M
The contents of accumulator wnen the program counter reaches 0109H is

(A) 20H (B) 02H (C) 00H (D) FFH.

Q28. GATE ECE 2005

Open solution

Consider an 8085 microprocessor system

If in addition following code exists from 0109H onwards ORI 40H, ADD M What will be the result in the accumulator after the last instruction is executed

(A) 40H (B) 20H (C) 60H (D) 42H

Q29. GATE ECE 2004

Open solution

The number of memory cycles required to execute the following 8085 instructions
(I.) LDA 3000H
(II.) LXI D, FOF 1H
Would be

(A) 2 for (I) and 2 for (II) (B) 4 for (I)and 3 for (II)
(C) 3 for (I)and 3 for (II) (D) 3 for (I) and 4 for (II)

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Q30. GATE ECE 2004

It is desired to multiply the numbers 0AH by 0BH and store the result in the accumulator. The numbers are available in registers B and C respectively. A part of the 8085 program for this purpose is given below:

MVI A, 00H
LOOP:.....
.....
HLT END

The sequence of instruction to be complete the program would be

(A) JNZ LOOP, ADD B, DCR C
(B) ADD B, JNZ LOOP, DCR C
(C) DCR C, JNZ LOOP, ADD B
(D) ADD B, DCR C, JNZ LOOP

Open solution

Q31. GATE ECE 2004

Consider the sequence of 8085 instructions given below.

LXI H, 9258
MOV A, M
CMA
MOV M, A

Which one of the following is performed by this sequence?

(A) contents of location 9258 are moved to the accumulator.
(B) contents of location 9258 are compared with the contents of the accumulator.
(C) contents of location 9258 are complemented and stored in location 9258.
(D) contents of location 5892 are complemented and stored in location 5892.

Open solution

Q32. GATE ECE 2003

In an 8085 microprocessor, the instruction CMP B has been executed while the content of the accumulator is less than that of register B. As a result

(A) Carry flag will be set but Zero flag will be reset
(B) Carry flag will be reset but Zero flag will be set
(C) Both Carry flag and Zero flag will be reset
(D) Both Carry flag and Zero flag will be set

Open solution

Q33. GATE ECE 2002

The contents of Register (B) and Accumulator (A) of 8085 microprocessor are 49H and 3AH respectively. The contents of A and the status of carry flag (CY) and sign flag (S) after executing SUB B instructions are

(A) A = F1, CY = 1, S =1
(B) A = 0F, CY = 1, S =1
(C) A = F0, CY = 0, S =0
(D) A = 1F, CY = 1, S =1

Open solution

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Q34. GATE ECE 2002

Open solution

Consider the following assembly language program.

```
        MVI     B,87H
        MOV     A,B
START:  JMP     NEXT
        MVI     B,00H
        XRA     B
        OUT     PORT1
        HLT
NEXT:   XRA     B
        JP      START
        OUT     PORT2
        HLT
```

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The execution of the above program in an 8085 microprocessor will result in

(A) an output of 87H at port 1

(B) an output of 87H at port 2

(C) infinite looping of the program execution with accumulator data remaining at 00H.

(D) infinite looping of the program execution with accumulator data alternating between 00H and 87H

Marks 5 2 questions

Q35. GATE ECE 1994

Open solution

Write down the sequence of Instructions which are actually (till HLT instruction), if the program begins with the location 1FF5H.

Address (HEX)	8085 Instructions
1FF5	XRA A
1FF6	LXI H, 200H
1FF9	PCHL
1FFA	HCT
1FFB	LXI H, 2100H
1FFE	ANI 00
2000	LXI H, FFFFH
2003	INX H
2004	JZ 2100H
2007	HLT
2100	LXI H, IFFFH
2103	MOV A, M
2104	INR A
2105	HLT

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Q36. GATE ECE 1991

Open solution

The program given below is run on an 8085 based microcomputer system. Determine the contents of the registers: PC, SP, B, C, H, L after a half instruction is executed.

```
LOC
2000 START: LXI SP, 1000H
LXI H, 2F37 H
XRA A
MOV A, H
INX H
PUSH H
CZ 20 FF H JMP 3000 H
HLT
20FF ADD H
RZ
POP B
PUSH B
RNZ
HLT
```

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Answer Key

Marks 1

1. **14**	2. **C**	3. **D**	4. **C**	5. **D**
6. **B**	7. **A**	8. **C**	9. **C**	10. **C**
11. **D**	12. **C**	13. **B**	14. **PC= 200 AH** **SP = OFF EH**	15. **A**
16. **A**				

Marks 2

17. **A**	18. **B**	19. **C**	20. **A**	21. **C**
22. **C**	23. **C**	24. **B**	25. **B**	26. **B**
27. **C**	28. **C**	29. **B**	30. **D**	31. **C**
32. **A**	33. **A**	34. **B**		

Marks 5

35. **Solve it.**	36. **PC = 3001 H** **SP = OFFE H** **BC = 200D H** **HL = 2F38 H**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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